Samuel Degen

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Education

University of California Los Angeles (UCLA) *9 / 2023 - present*
B.S. Physics – GPA: 3.986 (overall); 4.0 (major); 15 graduate physics classes
Advisors: Zvi Bern & Mikhail Solon (Scattering Amplitudes)

University of Colorado Boulder *2021 - 2023*
High School Concurrent Enrollment – GPA: 4.0 (overall)

Publications

1. S. Wang, **S. Degen,** and H. Liang, "Benchmarking FRG-DFT Against Exact Thermodynamics for the Single-site Bose-Hubbard Model", manuscript **in preparation**, 2026.

Honors

UCLA URC-Sciences Summer Program (URCSSP) *Summer 2026*
Undergraduate Research Center - Sciences, UCLA; Los Angeles, CA, USA
Advisors: Mikhail Solon & Zvi Bern (Scattering Amplitudes)

- Selected from 130+ applicants for fully-funded 10-week research fellowship, supporting the proposal "Precision Modeling of Environmental Effects in Gravitational Wave Signatures"

Goldwater Scholarship *2026 - 2027*
Barry Goldwater Scholarship and Excellence in Education Foundation

- National undergraduate research scholarship recognizing exceptional promise for a research career in the natural sciences, mathematics, or engineering
- 1 of 454 students nationwide selected; 1 of 2 UCLA students selected in 2026

APS Steven Chu Award for Best Undergraduate Research (1st) *10 / 2025*
2025 Annual Meeting of the Far West Section of the American Physical Society (APS); UC Santa Cruz, CA, USA

- Awarded 1st Prize for the talk "Towards a Nonperturbative Many-Body Nuclear Theory: Benchmarking FRG-DFT with the One-Dimensional Fermi Gas"
- Selected as the top undergraduate research talk (theory or experiment) among presenters from California, Nevada, and Hawaii.

UCLA Undergraduate Research Scholars Program (URSP) *2025 - 2026*
Undergraduate Research Center - Sciences, UCLA; Los Angeles, CA, USA

- 1 of 3 physics students selected for UCLA's most prestigious undergraduate research scholarship, supporting the proposal "Effects of Black Hole Environments on Gravitational Wave Signals: A Field-Theoretic Approach" with Mikhail Solon

Mani L. Bhaumik Institute for Theoretical Physics Summer Research Fellowship *8 - 9 / 2025*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA; Los Angeles, CA, USA
Advisors: Zvi Bern & Mikhail Solon (Scattering Amplitudes)

- First undergraduate offered a summer research fellowship at UCLA's Bhaumik Institute for Theoretical Physics

University of Tokyo Research Internship Program (UTRIP) *6 - 7 / 2025*

Quark Nuclear Science Institute, The University of Tokyo (東京大学)*; Tokyo, Japan*

Advisor: Haozhao Liang (Nuclear Theory)

- 1 of 14 students selected from 1,149 applicants for in-person, fully funded 6-week research and culture program in Tokyo
- *Friends of UTokyo, Inc. (FUTI) Global Leadership Award* – prestigious funding award for USA students to research in Japan (1 of 2 UTRIP students)

Perimeter Institute's PSI START Satellite Program *5 / 2025*

(Perimeter Scholars International Students' Training Accelerator for Research in Theory)

Bishop's University; Sherbrooke, Quebec, Canada

- 1 of 3 international students selected for in-person, fully funded 2-week intensive theoretical physics coursework in Canada

UCLA Summer Undergraduate Research Fellowship *Summer 2024*

Dept. of Physics & Astronomy, UCLA; Los Angeles, CA, USA

Advisor: E. Paulo Alves (Plasma Theory)

- 1 of 2 first-year undergraduates selected for competitive paid 10-week research fellowship

(declined) **USTC Future Scientist Exchange Program (FuSEP)** *6 - 7 / 2026*

University of Science and Technology of China (中国科学技术大学)*; Hefei, China*

- Selected for in-person, fully funded 6-week research and culture program in Hefei. Offered for Nuclear Theory or String Theory.

Presenting & Talks

◊ = **invited**
★ = **awarded**

Research

"Dynamical Friction from Phonon Emission Amplitudes"

- GWSky Workshop, "Precision Gravitational Wave Astronomy: From Theory to Discovery", Sexten Center for Astrophysics Riccardo Giacconi; Sexten, Italy *[planned July 27-31, 2026]*

"Gravitational Drag from Phonon Emission"

- Amplitudes Journal Club, UCLA *April 23, 2026*
- California Amplitudes Meeting 2026 (Davis), UC Davis *[planned May 30, 2026]*

"Effects of Black Hole Environments on Gravitational Wave Signals"

- California Amplitudes Meeting 2025 (LA), UCLA *November 8, 2025*
- Undergraduate Research & Creativity Showcase, UCLA *[planned May 19, 2026]*

"Towards a Nonperturbative Many-Body Nuclear Theory: Benchmarking FRG-DFT with the One-Dimensional Fermi Gas"

- UTRIP Final Presentation, The University of Tokyo *August 5, 2025*
- ★ Nuclear Physics Oral Contributions (L04.00005), APS Far West Section Meeting, UC Santa Cruz *October 11, 2025*
  - *Steven Chu Award (1st)*
- Nuclear Theory Oral Contributions, APS Global Physics Summit 2026; Denver, Colorado *March 17, 2026*

"Physics Without Borders: Nuclear Theory and Global Perspectives from a Summer in Tokyo"

- ◊ Special Seminar, Physics Summer Research Experience for Undergraduates (REU), UCLA *August 15, 2025*

"Data-driven statistical model of nonthermal particle acceleration by the kink instability in relativistic jets"

- Department of Physics Summer Research Talks, UCLA — *August 22, 2024*
- Plasma Astrophysics Oral Contributions (NO05.00014), 66th Annual Meeting of the APS Division of Plasma Physics; Atlanta, Georgia — *October 9, 2024*

Pedagogy

"An Introduction to the Dark Photon"

- PHYS 226D Final Presentation, UCLA — *December 4, 2025*

"From Finite Groups to Lie Algebras: Symmetries and Representations in Physics"

- Perimeter Institute's PSI START Satellite Program, Bishop's University; Quebec, Canada — *May 29, 2025*

"Convex Compactness and its Applications"

- APPM 6560 Final Presentation, CU Boulder — *May 1, 2023*

Meetings & Schools

GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery — *[planned 7 / 2026]*
Sexten Center for Astrophysics Riccardo Giacconi, Italy

2026 IHES Summer School - Cosmological Correlators — *[planned 7 / 2026]*
Institut des Hautes Études Scientifiques (IHES), France
(Selected participant; advanced international school primarily for graduate students and postdocs)

California Amplitudes Meeting 2026 (Davis) — *[planned 5 / 2026]*
Center for Quantum Mathematics and Physics (QMAP), UC Davis

APS Global Physics Summit 2026 — *3 / 2026*
Denver, Colorado

Accelerating Math and Theoretical Physics with AI — *3 / 2026*
Institute for Pure and Applied Mathematics (IPAM), UCLA

GWSky Kick-off Meeting — *12 / 2025*
Max Planck Institute for Gravitational Physics (Albert Einstein Institute–AEI), Potsdam, Germany

California Amplitudes Meeting 2025 (LA) — *11 / 2025*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA

2025 Annual Meeting of the APS Far West Section — *10 / 2025*
UC Santa Cruz

2025 Southern California Strings — *10 / 2025*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA

California Amplitudes Meeting 2025 (Davis) — *5 / 2025*
Center for Quantum Mathematics and Physics (QMAP), UC Davis

66th Annual Meeting of the APS Division of Plasma Physics — *10 / 2024*
Atlanta, Georgia

Research Experience

UCLA – Scattering Amplitudes & Gravitational Physics — *8 / 2025 - present*
Advisors: Zvi Bern & Mikhail Solon

- Developing a field-theoretic framework for environmental effects in gravitational-wave systems, with the goal of computing how surrounding media such as gas or dark matter modify compact-object dynamics.
- Formulating gravitational dynamical friction in hydrodynamic effective field theory as radiation into on-shell phonon modes, connecting classical drag forces to scattering-amplitude methods.

- Computing higher-order corrections to the leading dynamical-friction force using perturbative QFT techniques, with attention to graviton–phonon interactions, gauge invariance, and Ward identities in a medium that breaks Lorentz symmetry.
- Drawing conceptual and technical connections between gravitational environmental effects and jet-quenching physics in QCD, where real-time propagation through a medium produces analogous dissipative phenomena.
- Youngest member and only undergraduate in the European Research Council's GWSky collaboration, contributing to a new international effort on precision theory for next-generation gravitational-wave astronomy
- Presenting this work at the GWSky Workshop, California Amplitudes Meeting, & UCLA Amplitudes Journal Club.

Supported by UCLA URCSSP, Mani L. Bhaumik Institute Summer Research Fellowship, and UCLA URSP.

The University of Tokyo (東京大学) – Nuclear Theory *5 / 2025 - present*

Advisor: Haozhao Liang

- Applying functional renormalization group-aided density functional theory (FRG-DFT) to exactly solvable quantum many-body systems, with the goal of benchmarking nonperturbative first-principles approaches to nuclear density functionals.
- Deriving and truncating flow-equation hierarchies for ground-state energies and correlation functions, enabling controlled comparisons between FRG-DFT, perturbation theory, and exact thermodynamics.
- Developing numerical implementations to solve coupled flow equations and test the accuracy of approximate truncations across interaction strengths and densities.
- Co-authoring a manuscript on benchmarking FRG-DFT against exact thermodynamics for the single-site Bose-Hubbard model, with broader applications to strongly correlated fermions and nuclear many-body theory.
- Presented related work at the University of Tokyo, APS Far West Section Meeting, and APS Global Physics Summit 2026; awarded 1st prize of the APS Steven Chu Award for Best Undergraduate Research for this project.

Supported by UTRIP 2025 and FUTI Global Leadership Award.

UCLA – Plasma Theory *4 / 2024 - 1 / 2025*

Advisor: E. Paulo Alves

- Developed analytic and machine-learning methods to study time-dependent nonthermal particle acceleration in relativistic astrophysical jets.
- Formulated an inverse-modeling approach to distinguish physical acceleration histories from degenerate solutions that reproduce the same observed particle spectrum.
- Demonstrated limitations of simple energy-dependent acceleration models and identified more flexible time-dependent descriptions consistent with simulated jet dynamics.
- Presented results in an oral contribution at the 66th Annual Meeting of the APS Division of Plasma Physics.

Supported by UCLA Physics Summer 2024 Research Fellowship.

UCLA – Condensed Matter *1/2024 - 3/2024*

Advisor: B.C. Regan

- Developed an undergraduate-level pedagogical treatment of the Feynman checkerboard model, emphasizing its connection to propagators and relativistic quantum mechanics.

Relevant Courses

Final Papers

S. Degen, "An Introduction to 3D Gravity from TQFTs" *6 / 2025*

(UCLA PHYS 242C Final Project)

S. Degen, "Tensor Products of Representations of Lie groups" (UCLA MATH 229A Final Project) *10 / 2024*

UCLA	Graduate	PHYS 215A	Statistical Physics
* *Spring 2026*		PHYS 221A	Quantum Mechanics I
† *final papers linked above*		PHYS 221B	Quantum Mechanics II
		PHYS 226B	Particle Physics (Standard Model)
		PHYS 226C	Particle Physics (QCD and Higgs)
		PHYS 226D	Beyond the Standard Model
		PHYS 230A	Quantum Field Theory I
		PHYS 230B	Quantum Field Theory II
		PHYS 230C	Quantum Field Theory III
		PHYS 230D*	Advanced Quantum Field Theory
		PHYS 231B	Mathematical Physics (Lie Theory)
		PHYS 232B	Gravitational Waves & The Double Copy
		PHYS 242C†	Topological Quantum Field Theory
		PHYS 291	String Theory Journal Club
		PHYS 495	Teaching College Physics
		MATH 229A†	Lie Groups and Lie Algebras
	Undergrad	PHYS 140A	Solid State
		PHYS 199*	Directed Research
		CSCI 174A	Computer Graphics (JavaScript)
		JAPAN 1-3	Elementary Modern Japanese
		CHIN 1-2	Elementary Modern Chinese
CU Boulder	Graduate	PHYS 5770	Gravitational Theory
		APPM 6560	Measure-Theoretic Probability
	Undergrad	CSCI 3104	Algorithms (C++)
		CSCI 4622	Machine Learning (Python)

Community & Service

Peer Mentor, Division of Physical Sciences, UCLA *2025 - present*

- Provided first-year physical science students at UCLA guidance on coursework, professional development, fellowships, and early research opportunities, especially for those interested in theoretical physics

Contributor, Amplitudes Journal Club, UCLA Department of Physics *1 / 2025 - present*

- First undergraduate member and presenter of this journal club

Member, String Theory Journal Club, UCLA Department of Physics *1 / 2025 - present*

- First undergraduate member of this journal club

Professional Events Chair, Society of Physics Students, UCLA *2025*

- Developed extensive resources for summer research opportunities and course planning

Contributor, Plasma Theory Journal Club, UCLA Department of Physics *4 / 2024 - 1 / 2025*

- Multiple presentations and analyses of recent high-impact papers to theoretical plasma PhD students and faculty
- Discussed theory and application of novel analytic and machine learning methods presented in the club to current projects in the UCLA Plasma Theory group
- First undergraduate to present in this journal club

Interviews & Press

Goldwater Scholarship (Scholars List) (UCLA College Interview) (UCLA Physics News) *3 / 2026*

APS Steven Chu Award for Best Undergraduate Research (APS website) (UCLA Physics News) (APS FWS Newsletter 2025, pg. 6) *10 / 2025*

FUTI Global Leadership Award (report) *8 / 2025*

University of Tokyo Research Internship Program (program overview) (final report) *6 / 2025*

Perimeter Institute's PSI START Satellite (announcement) (interview) *5 / 2025*